Batterjee Medical College
For Science and Technology

Approach to a case of Heart burn

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Gastroesophageal Reflux Disease

Learning Objectives

Knowledge & understanding

- Explain the mechanisms and risk factors contributing to gastroesophageal junction dysfunction and impaired esophageal acid clearance in GERD.
- Describe the typical, atypical, and extraesophageal clinical manifestations of GERD.

Skill

- Apply a stepwise diagnostic approach to a patient with suspected GERD, determining when an empiric PPI trial versus upfront endoscopy is indicated.
- Formulate an evidence-based management plan for a patient with GERD, incorporating lifestyle modification, pharmacotherapy, and surgical referral criteria

Main subtopics

- Definition and Pathophysiology of GERD
- Etiology and Risk Factors
- Clinical Features
- Differential Diagnosis
- Diagnostics
- Treatment
- Barrett Esophagus
- Complications

Case scenario :

A 58-year-old man comes to the physician with a 12-year history of recurrent heartburn. He has also had a cough, which is worse at night, for the past year. He has not seen a physician for 8 years. He has smoked one pack of cigarettes daily for 30 years. His only medication is an over-the-counter antacid. He is 175 cm tall and weighs 95 kg; BMI is 31 kg/m2. Vital signs are within normal limits. Physical examination shows no lymphadenopathy. The abdomen is soft and nontender. The remainder of the examination shows no abnormalities. Results of a complete blood count are within the reference ranges. An upper endoscopy shows dark pink mucosa up to 2 cm proximal to the gastroesophageal junction. Biopsy samples of the mucosa show columnar epithelium with goblet cells and low-grade dysplasia.

Case problem :

Which of the following is the most appropriate next step in management?

A. Repeat endoscopy in 24 months

B. Endoscopic radiofrequency ablation

C. Esophagectomy

D. Omeprazole, clarithromycin, and metronidazole

E. External beam radiation therapy

F. Nissen fundoplication

Gastroesophageal reflux disease

Gastroesophageal reflux: regurgitation of stomach contents into the esophagus

Gastroesophageal reflux disease (GERD)

A condition in which reflux causes troublesome symptoms (typically including heartburn or regurgitation) and/or esophageal injury/complications

Etiology

Mechanisms

Gastroesophageal junction dysfunction :

- Increased frequency of transient lower esophageal sphincter (LES) relaxations
- Imbalance between intragastric and LES pressures
  - LES tone can be decreased by substances such as caffeine and nitroglycerin
  - Intragastric pressure is increased:
    - Pregnancy
    - Delayed gastric emptying
    - Obesity
- Anatomic abnormalities of gastroesophageal junction

Impaired esophageal acid clearance

- Disrupted by reduced salivation (e.g., due to smoking)
- Decreased peristalsis

Etiology

Risk factors for GERD

- Smoking
- Caffeine
- Alcohol consumption
- Stress
- Obesity
- Pregnancy
- Scleroderma
- Sliding hiatal hernia: ≥ 90% of patients with severe GERD
- Asthma

Clinical features

Typical symptoms

- Retrosternal burning pain (heartburn)
- Regurgitation
- Dysphagia

Atypical symptoms

- Pressure sensation in the chest
- Belching
- Features of aspiration pneumonia

Clinical features

Extraesophageal symptoms

- Chronic nonproductive cough
- Hoarseness

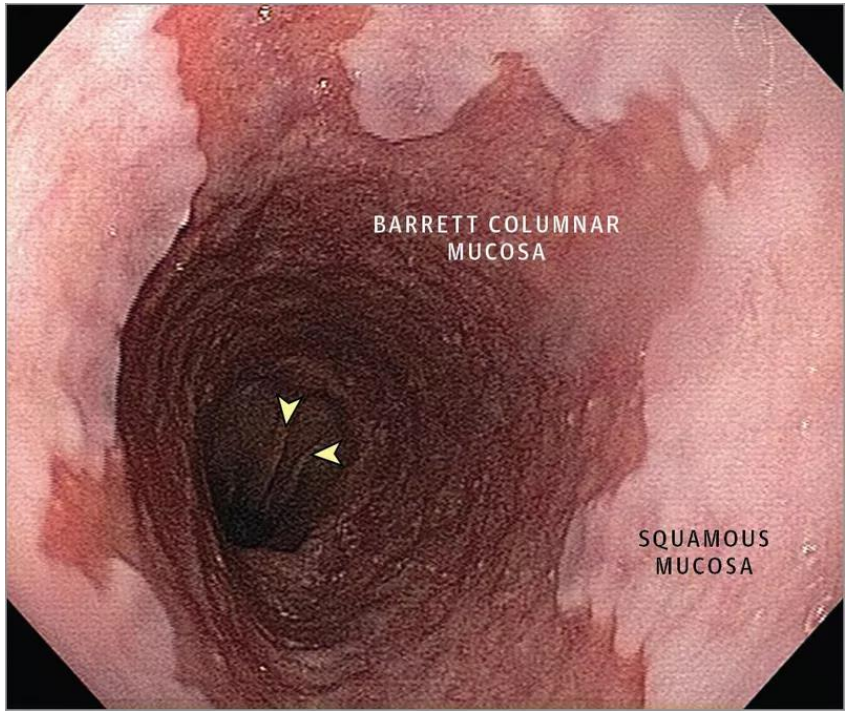
Aggravating factors

- Lying down shortly after meals
- Certain foods/beverages

Clinical features

Red flags in GERD

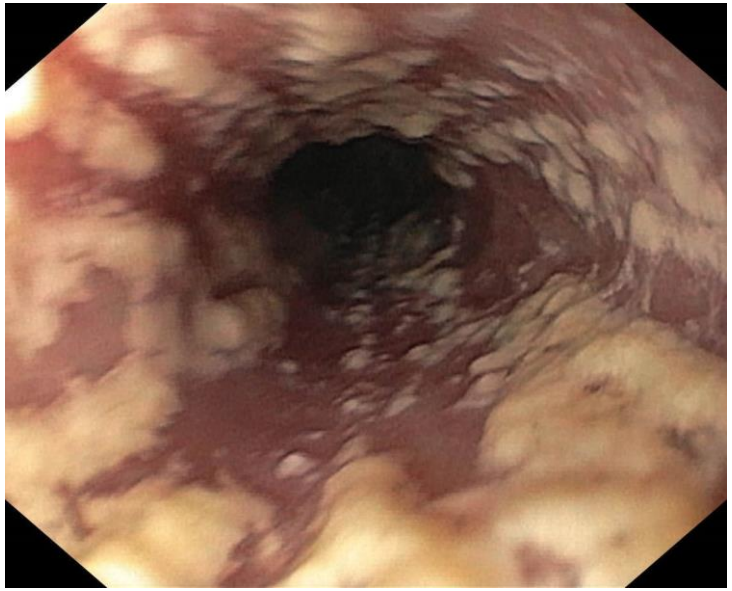
- Dysphagia, odynophagia
- Anemia and/or evidence of GI bleeding
- Unintentional weight loss
- Vomiting
- Signs of GI bleeding
- Presence of > 1 risk factor for Barrett esophagus:
  - **Male sex**
  - **European descent**
  - **Age ≥ 50 years**
  - **Obesity**
  - **Symptoms ≥ 5 years**



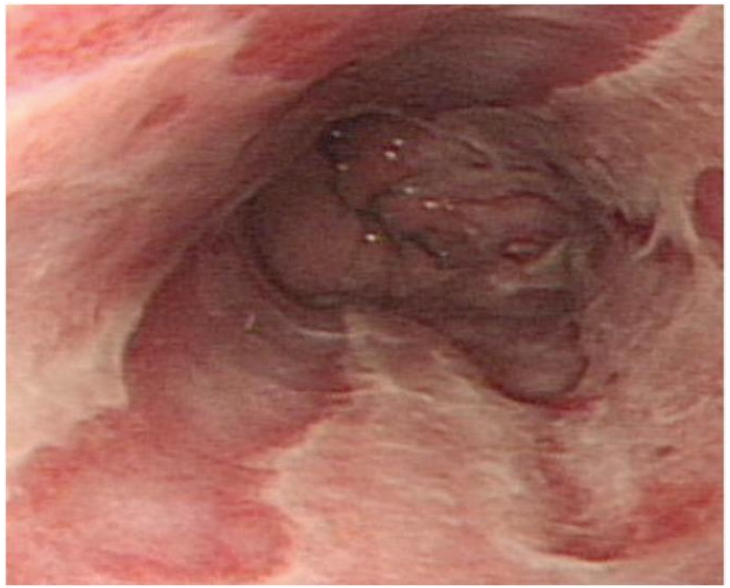
Differential diagnoses

Esophagitis

- Infectious esophagitis (typically seen in immunocompromised patients)
  - Esophageal candidiasis: white or yellow adherent plaques (pseudomembranes).
  - Herpes esophagitis: (mainly HSV-1); punched-out ulcers
  - CMV esophagitis: linear ulcers







Diagnostics

Approach

All patients

Rule out life-threatening differential diagnoses of GERD and chest pain e.g., (ACS)

Typical symptoms without red flags in GERD:

- Initiate treatment for GERD; start an empiric once-daily PPI trial.
- If there is relief after 8 weeks of PPI: GERD is likely

Diagnostics

Red flags in GERD:

Refer to gastroenterology for EGD before initiating treatment.

EGD Indications

- Red flags in GERD
- Risk factors for Barrett esophagus
- No symptomatic improvement after PPI trial

Esophageal pH monitoring

Not a routine diagnostic test

Treatment

Pharmacological therapy

PPIs: standard dose of PPI

Lifestyle changes

- Physical recommendations
- Weight loss in patients with obesity
- Elevate the head of the bed
- Reduce or avoid triggering substances
  - ✓ Tobacco
  - ✓ Alcohol
  - ✓ Caffeine

Treatment

Surgical therapy

Fundoplication

Indications

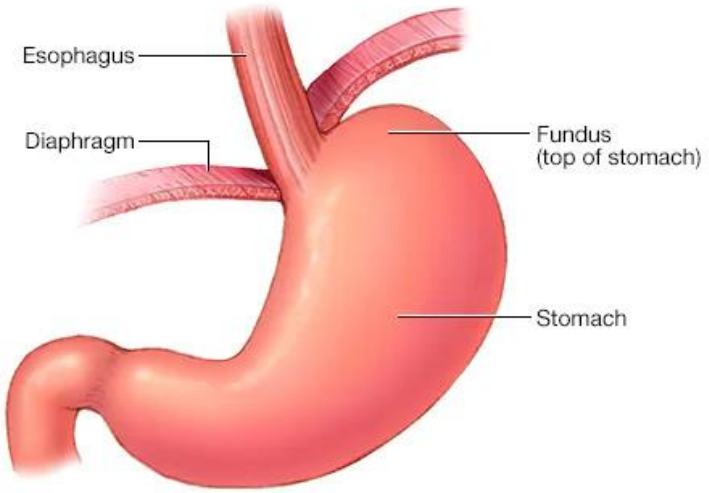
severe esophagitis, strictures, recurrent aspiration

Techniques

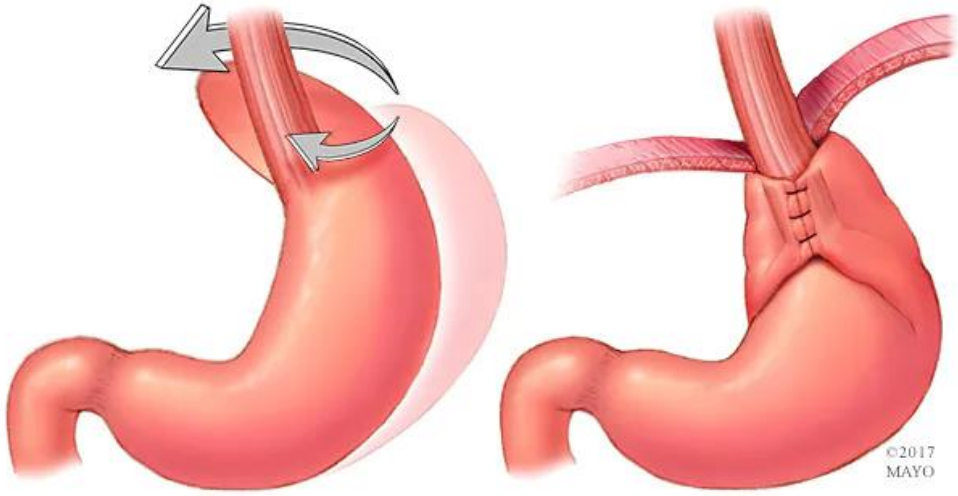
Complete fundoplication (Nissen fundoplication)

Considerations for patients with comorbidities

Patients with obesity and reflux undergoing bariatric surgery: Consider Roux-en-Y.



Nissen fundoplication



A. Fundus wrapped around back side of esophagus

B. Wrap secured with sutures to anchor lower esophagus below diaphragm

Complications

Barrett esophagus

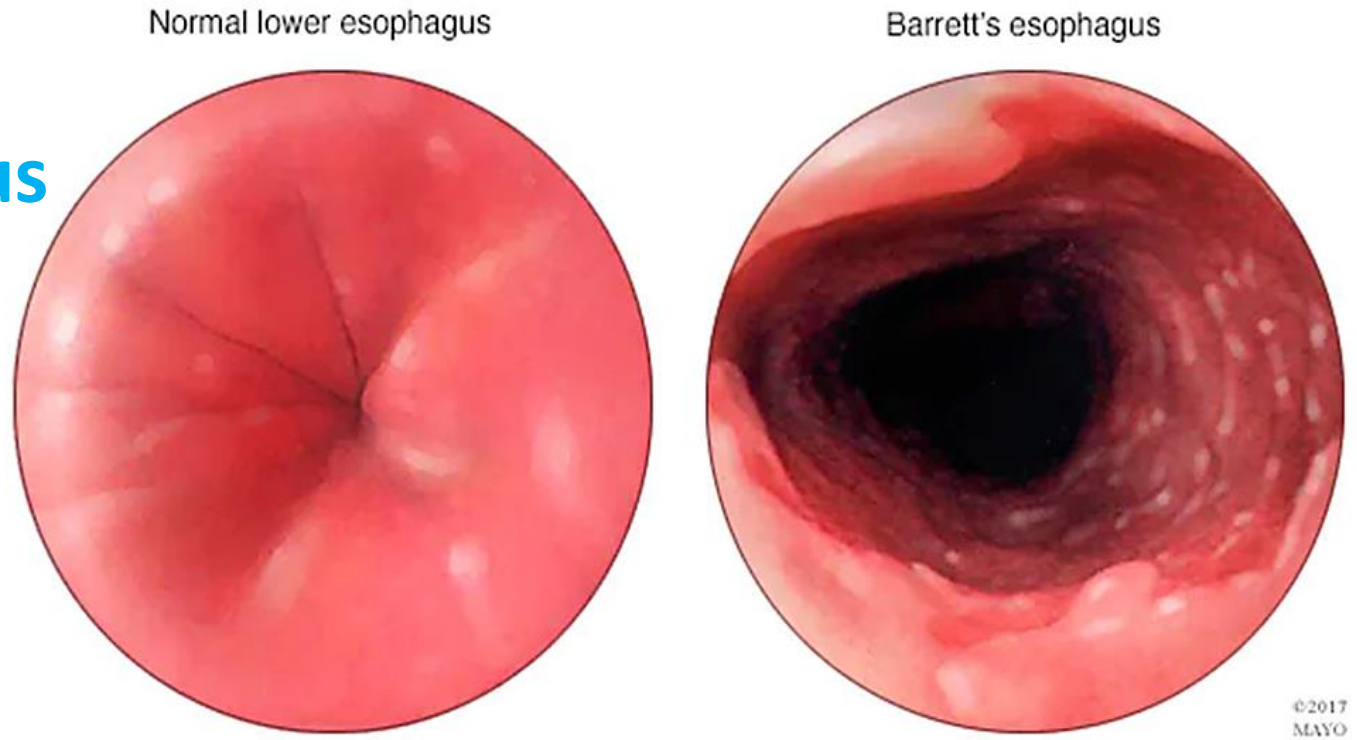
Definition:

Intestinal metaplasia of the esophageal mucosa induced by chronic reflux.

- Columnar epithelium instead of the normal squamous epithelium
- A premalignant change

Risk factors for Barrett esophagus

- Male sex
- European descent
- Age ≥ 50 years
- Obesity
- Symptoms ≥ 5 years



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Complications

Barrett esophagus

Pathophysiology

Reflux esophagitis

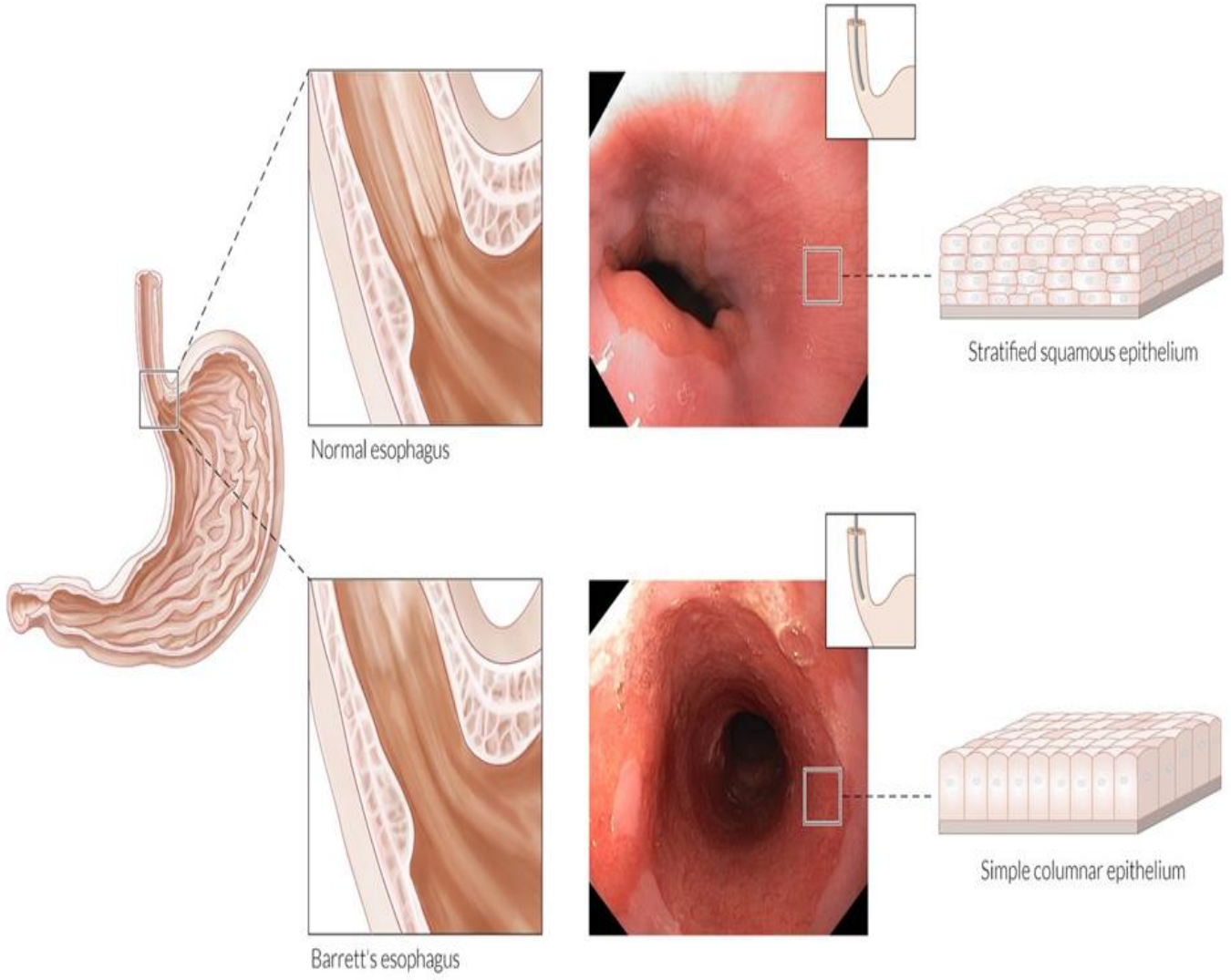
→ stomach acid damages mucosa of distal esophagus

→ nonkeratinized stratified squamous epithelium

→nonciliated columnar epithelium and goblet cells

= (intestinal metaplasia, Barrett metaplasia)

Complications: esophageal adenocarcinoma



Barrett esophagus

Management and surveillance

Endoscopy with four-quadrant biopsies at every 2 cm of the suspicious area (salmon-colored mucosa)

If high-grade dysplasia: malignant transformation (7% per year)

endoscopic treatment of mucosal irregularities

endoscopic eradication of dysplasia via

- Mucosal resection and/or
- Radiofrequency ablation

If low-grade dysplasia: malignant transformation (0.7% per year)

- Endoscopic eradication
- surveillance endoscopy every 12 months is an acceptable alternative.

Additional complications

Iron deficiency anemia:

Mucosal erosions and ulcerations → chronic bleeding → anemia

Esophageal stricture

Most common sequela of reflux esophagitis

Diagnostics

- Barium esophagram (best initial test)
- Endoscopy with biopsies: to rule out malignancy

Treatment

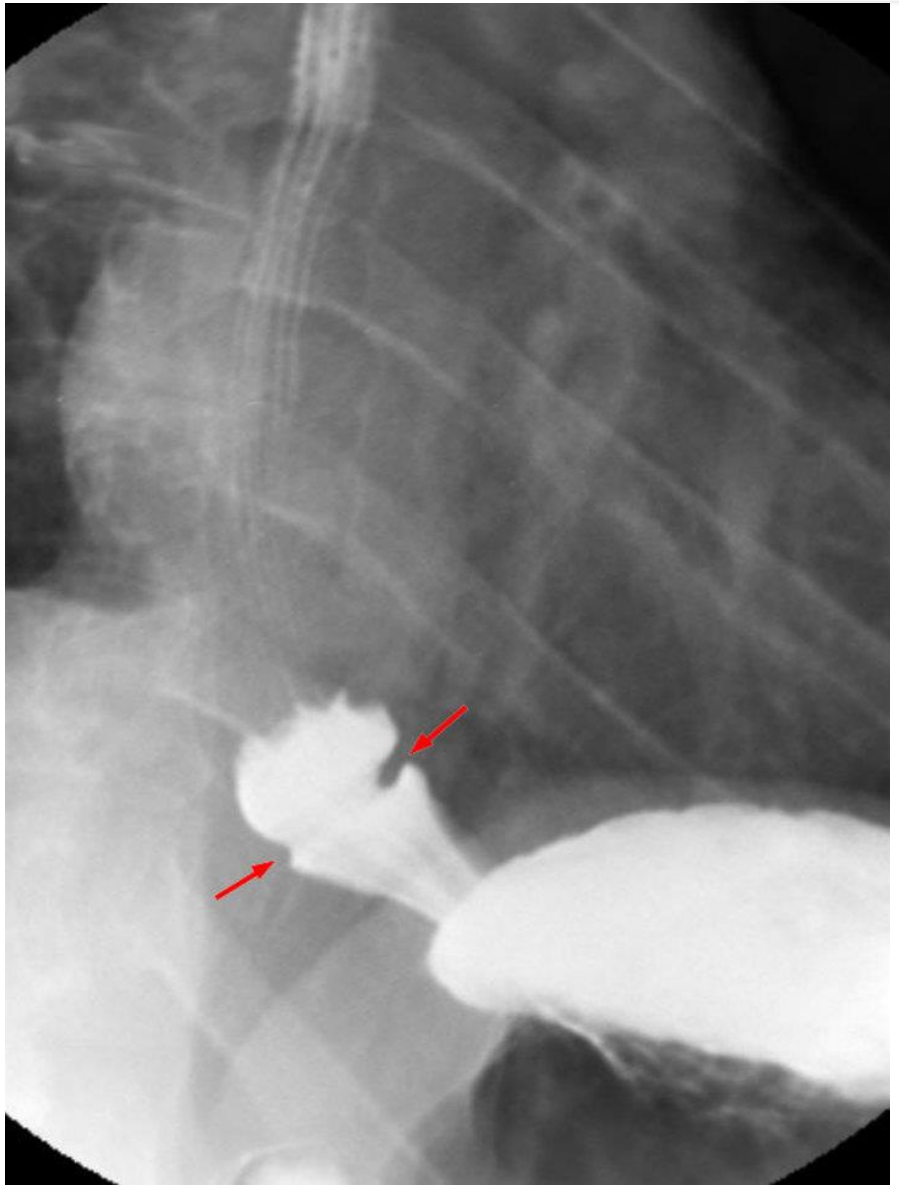
Dilation with bougie dilator/balloon dilator



Additional complications

Esophageal ring

- Schatzki rings: narrowing of the esophagus
  - Most commonly seen at the squamocolumnar junction
  - Usually caused by chronic acid reflux
  - Can lead to dysphagia
  - Usually treated with esophageal dilation



Additional complications

Complications due to aspiration of gastric contents;

- Aspiration pneumonia
- Chronic bronchitis
- Asthma
- Reflux laryngitis: hoarseness

References & further reading

- AMBOSS
- Harrison’s Principles of Internal Medicine 20th edition ISBN-13: 978-1259644030

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Thank You